DTSC Project 2 Part A - Credit Card Default Prediction Using Machine Learning

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1. Introduction

This analysis builds a classification model to predict whether a credit card customer will default on their next month's payment, using a dataset of 30,000 customers with information on credit limits, repayment history, bill statements, payment amounts, and demographic characteristics including age, gender, education, and marital status. Because identifying high-risk customers has direct financial implications, since missed defaults are typically more costly than false alarms, the objective is to develop a model that is both sensitive to true defaulters and practical for deployment in credit risk monitoring. To strengthen the predictive signal, ten engineered features were created to summarize customer behavior, including average credit utilization, payment-to-bill ratios, delinquency frequency, and recent changes in billing and payment activity, all of which help capture evolving customer risk behavior.

2. Model Selection Process

We define “best” as the model that achieves the strongest balance between identifying true defaulters and maintaining overall discriminative performance, prioritizing ROC-AUC over raw accuracy due to class imbalance, since accuracy can be misleading in datasets where the majority class dominates. Three models were evaluated on a held-out 20% test set: logistic regression as an interpretable baseline, a decision tree to capture nonlinear structure, and a random forest to reduce variance through ensembling. Additional models, including LDA and QDA, were explored but underperformed, likely due to violations of their underlying assumptions, particularly multicollinearity and the lack of normally distributed predictors in the dataset. A classification threshold of 0.5 was used to convert predicted probabilities into class labels. While this provides a reasonable baseline, lowering the threshold would increase recall if minimizing missed defaults becomes a priority. This highlights that model deployment decisions depend not only on predictive performance but also on the organization’s risk tolerance. The random forest

consistently performed best across the key evaluation metrics used to define model quality, achieving a ROC-AUC of 0.7785 and balanced accuracy of 0.7152, outperforming the decision tree (ROC-AUC: 0.7634, balanced accuracy: 0.7001) and logistic regression (ROC-AUC: 0.7477, balanced accuracy: 0.6926).

3. Final Model

The final model is a random forest classifier (200 trees, max depth 10, balanced class weights), selected for its superior out-of-sample performance and ability to capture complex interactions in customer behavior. On the test set, the model correctly identified 809 true defaulters out of 1,327 actual defaults (recall ≈ 0.61), while predicting 1,647 customers as high-risk, of which 809 were true positives. This reflects a practical tradeoff: the model captures a majority of default cases while maintaining a manageable level of false positives. Feature importance analysis indicates that recent repayment behavior is the dominant driver of predictions, with months late, maximum delinquency, and average delinquency among the most influential variables, followed by recent repayment status (PAY_0). These results align with domain expectations, suggesting that deteriorating payment patterns are strong indicators of default risk. Overall, the model provides a reliable and actionable approach for identifying high-risk customers and can support targeted intervention strategies such as prioritizing account review, monitoring high-risk customers, or initiating early outreach to mitigate potential defaults.

Appendix:

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DATA SHAPE:
(30000, 25)

NEW FEATURES CREATED:
['avg_bill_amt', 'avg_pay_amt', 'utilization_mean', 'pay_to_bill_ratio', 'max_delinquency',
'avg_delinquency', 'months_late', 'recent_bill_change', 'recent_payment_change', 'limit_per_age']

MODEL COMPARISON TABLE:
                Model  ROC_AUC  Accuracy  Balanced_Accuracy  Recall_Default_1  Precision_Default_1  F1_Default_1   TN   FP  FN  TP
        Random Forest   0.7785    0.7740             0.7152            0.6096               0.4912        0.5440 3835  838 518 809
        Decision Tree   0.7634    0.7463             0.7001            0.6172               0.4468        0.5184 3659 1014 508 819
  Logistic Regression   0.7477    0.7438             0.6926            0.6006               0.4418        0.5091 3666 1007 530 797

SELECTED FINAL MODEL:
Random Forest

FINAL MODEL RESULTS:
          Model  ROC_AUC  Accuracy  Balanced_Accuracy  Recall_Default_1  Precision_Default_1  F1_Default_1   TN  FP  FN  TP
Random Forest   0.7785     0.774             0.7152            0.6096               0.4912         0.544 3835 838 518 809

FINAL MODEL CONFUSION MATRIX:
[[3835  838]
 [ 518  809]]
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TOP 15 RANDOM FOREST FEATURE IMPORTANCES:
                 feature  importance
       num__months_late      0.1502
               num__PAY_0      0.1081
   num__avg_delinquency      0.0976
   num__max_delinquency      0.0974
               num__PAY_2      0.0490
     num__limit_per_age      0.0327
       num__avg_pay_amt      0.0308
  num__utilization_mean      0.0283
               num__PAY_3      0.0281
num__pay_to_bill_ratio      0.0263
               num__PAY_4      0.0245
      num__avg_bill_amt      0.0235
          num__LIMIT_BAL      0.0234
          num__BILL_AMT1      0.0223
           num__PAY_AMT1      0.0209
```

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TOP 10 NEGATIVE LOGISTIC COEFFICIENTS:
                 feature    coef
               num__PAY_5 -0.1053
             num__PAY_AMT2 -0.1058
               num__PAY_2 -0.1206
               num__PAY_3 -0.1218
               num__PAY_6 -0.1294
               cat__SEX_2 -0.1398
          cat__MARRIAGE_2 -0.1404
num__pay_to_bill_ratio -0.1620
      num__limit_per_age -0.1761
        cat__EDUCATION_4 -0.9123
```

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TOP 10 POSITIVE LOGISTIC COEFFICIENTS:
                 feature   coef
       num__months_late 0.8340
   num__max_delinquency 0.3057
               num__PAY_0 0.1790
          num__BILL_AMT2 0.1398
num__utilization_mean 0.1050
          num__BILL_AMT3 0.0835
          num__BILL_AMT4 0.0424
       cat__EDUCATION_2 0.0352
           num__PAY_AMT5 0.0252
       num__avg_bill_amt 0.0140
```